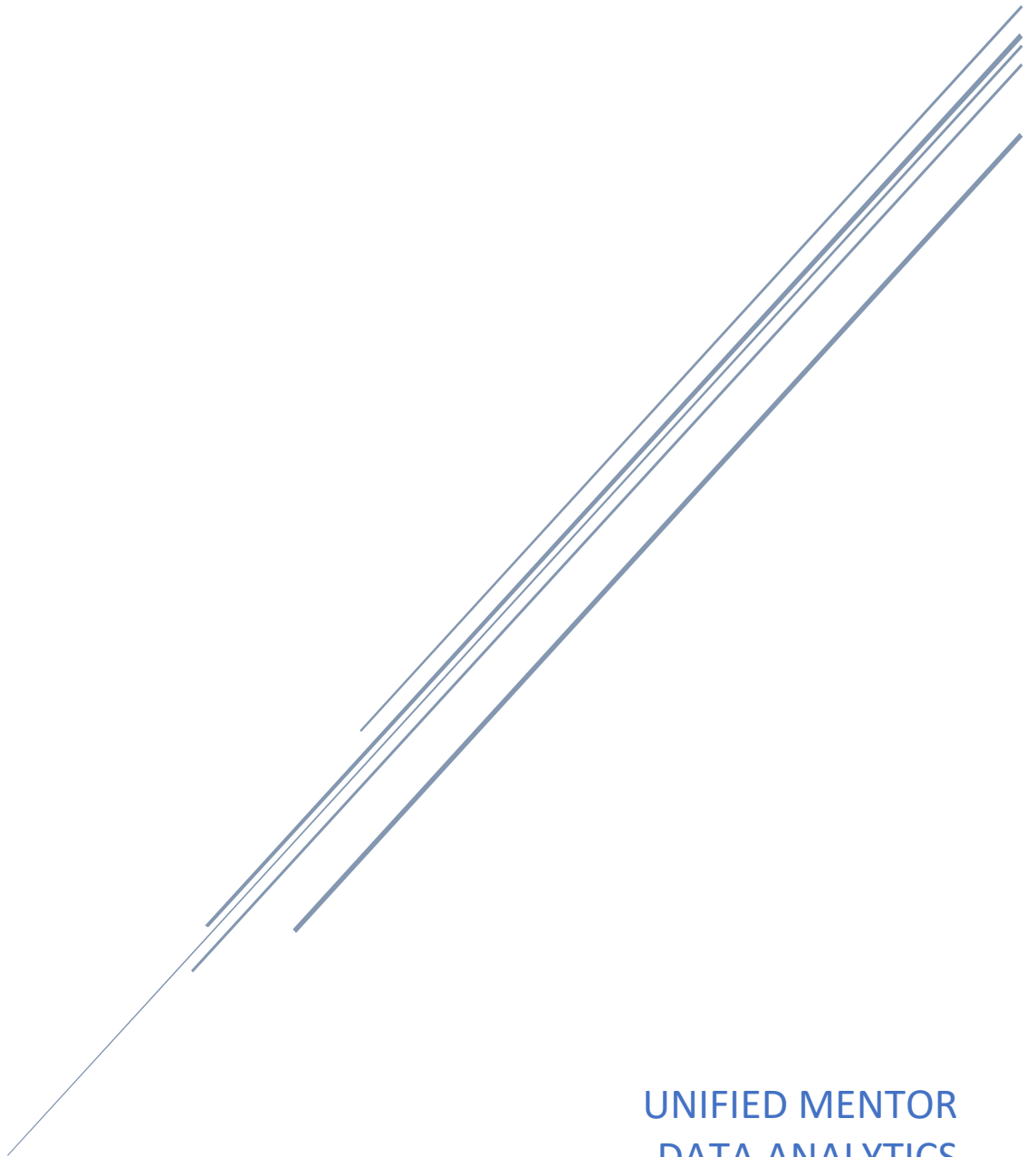


CUSTOMER SEGMENTATION AND CHURN PATTERN ANALYTICS IN EUROPEAN BANKING



UNIFIED MENTOR
DATA ANALYTICS

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Final report on
**“Customer Segmentation and Churn Pattern analytics
in European Banking”**

UNIFIED MENTOR

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ABSTRACT

The project titled “Customer Segmentation and Churn Pattern Analytics in European Banking” focuses on analysing customer behavior, identifying churn patterns, and developing an interactive analytical dashboard to support strategic decision-making in the banking sector. With increasing competition in the European Banking industry, customer retention has become one of the most critical business objectives. Banks continuously face challenges in understanding why customers discontinue services and how customer characteristics influence churn behavior. This project aims to address these challenges by applying data analytics, customer segmentation techniques, and visualization methods to derive meaningful business insights from banking customer data.

The project utilizes a cleaned banking dataset containing customer demographic, financial and behavioral attributes such as geography, age, credit score, account balance, salary, tenure, and activity status. Using Python libraries including Pandas, NumPy, Plotly, and Streamlit, the project performs exploratory data analysis and develops a professional dashboard that enables dynamic interaction with the data. The dashboard is designed to help banking professionals monitor customer churn trends, identify high-risk customer groups, evaluate customer engagement patterns effectively.

The dashboard in the project includes multiple interactive visualizations such as geography-wise churn analysis, age group churn trends, tenure-based churn comparison, and high-value customer churn exploration. Filters are implemented to allow users to analyse churn patterns across different customer segments dynamically. Additionally, a drill down customer view enables detailed inspection of churned and retained customers, improving analytical depth and usability for business users.

The implementation of this project demonstrates the practical application of data analytics and business intelligence tools in the financial sector. By combining data visualization with customer segmentation techniques, the dashboard provides a user-friendly and efficient platform for decision-making. The project can help banking institutions improve customer retention strategies, enhance customer satisfaction, and reduce financial losses associated with customer churn.

In conclusion, this project successfully integrates customer analytics, churn prediction concepts, and interactive dashboard development to deliver actionable insights into European banking customer behavior. The developed system serves as a valuable analytical solution for understanding customer segmentation and churn dynamics, ultimately supporting data-driven business strategies in the banking industry.

INTRODUCTION

1.1 About The Company

Unified Mentor, a global educational technology platform was established in 2022 with a mission to elevate employability through tailor-made skill development programs. Their offerings encompass a wide array of online courses, certification programs and internships designed to empower individuals in their quest and progress in the realms of technology, data, and marketing.

1.2 About The Project

The project is designed to analyze customer behavior and identify the major factors contributing to customer churn in the European banking sector. In today's competitive financial environment, retaining customer has become a significant challenge for banks, making customer analytics an essential part of business strategy. This project focuses on understanding customer segmentation patterns and churn trends through data analysis and interactive dashboard visualization.

The project is developed using Python and utilities important data analysis and visualization libraries such as Pandas, NumPy, Plotly, and Streamlit. A cleaned banking dataset containing customer information such as geography, age, credit score, account balance, tenure, salary, and activity status is used for the analysis. The Streamlit platform provides an interactive platform where users can filter and explore customer data based on multiple parameters including geography segment, age group, credit score, tenure group, and balance segment.

The dashboard includes several Key Performance Indicators (KPIs) such as Overall Churn Rate, High-Value Customer Churn Ratio, Geographic Risk Index, and Engagement Drop Indicator. These metrics help in measuring customer retention performance and identifying high risk customer groups. In addition, the project incorporates visual analytics such as geography-wise churn comparison, age group churn trends, tenure-based churn analysis, high-value customer exploration through scatter plots and bar charts.

The system also provides a drill-down customer view that allows detailed examination of churned and retained customers for better business understanding. By identifying inactive customers, premium customer churn behavior, and regional churn risks, the project supports data-driven decision-making for customer retention strategies.

Overall, this project demonstrates how customer segmentation and churn analysis can help banking institutions improve customer satisfaction, reduce customer attrition, and enhance long-term business performance through interactive business intelligence solutions.

1.3 Objectives And Deliverables

Objectives of the project:

The main objective of the project is to analyze customer behavior and understand the factors influencing customer churn in the banking sector. Using the Streamlit dashboard in

Python, the project provides interactive analytics and visual insights that help identify high-risk customer data such as geography, age group, credit score, account balance, tenure, and activity status to evaluate churn patterns across different customer segments.

A. Primary objectives

- **Measure overall churn rate**
Indicates the percentage of customers who have exited the bank. This KPI helps evaluate the bank's customer retention performance and highlights the overall risk of customer loss.
- **Identify churn distribution across customer segments**
The dashboard analyzes churn behavior among different age groups, balance categories, tenure groups, and credit score segments. This segmentation helps determine which types of customers are more likely to leave the bank.
- **Compare churn behavior across European regions**
By analyzing geography-wise churn patterns. Regional comparisons help identify locations with higher churn risks and provide insights into customer behavior different across European markets.

Additionally, the project focuses on identifying inactive customers, evaluating high-value customer churn, and providing drill-down customer analysis to support better business decision-making and customer retention strategies.

B. Secondary objectives

- **Understand churn among high-value customers**
The dashboard identifies premium customers based on account balance and analyzes their churn patterns. This helps banks recognise whether valuable customers are leaving and enables the development of personalized retention strategies for profitable customer segments.
- **Evaluate engagement and tenure patterns**
This project examines how customer activity status and years of association with the bank influence churn behavior.
- **Support strategic planning and marketing decisions**
By providing visual insights and KPI-based analysis. Interactive charts, segment analysis, and drill-down customer views help decision-makers identify risk areas, target specific customer groups, and create effective marketing and retention campaigns.

Overall, these secondary objectives enhance the analytical value of the project and contribute to data-driven decision-making in the banking sector.

Deliverables of the project:

The project delivers a fully functional and interactive **Streamlit-based customer churn analytics dashboard** for the European banking sector. The dashboard is designed to provide real-time analytical insights using customer data and dynamic visualizations.

One of the key deliverables is the **interactive filtering system**, where users can filter data based on geography segment, age group, credit score segment, tenure group, and balance segment. This feature improves analytical flexibility and allows detailed exploration of customer churn patterns.

The project also delivers a set of important **Key Performance Indicator (KPI)**, including Overall churn rate, High-value customer churn ratio, Geographic risk index, and Engagement drop indicator. These KPIs provide quick and effective insights into customer retention performance and business risks.

Another major deliverable is the collection of **interactive visualizations and charts** created using Plotly. These include geography-wise churn analysis, age group churn trends, tenure-based churn comparison, and high-value customer churn exploration. These visualizations simplify complex data and improve decision-making efficiency.

The project additionally provides a **segment churn analysis table** and a **drill down customer view**, allowing users to inspect churned and retained customers in detail. This improves business understanding and supports customer-focused strategies.

Finally, the project delivers a complete analytical solution that combines customer segmentation, churn analysis, business intelligence, and dashboard development into a user-friendly system that can support banking organizations in improving customer retention and strategic decision-making.

METHODOLOGY

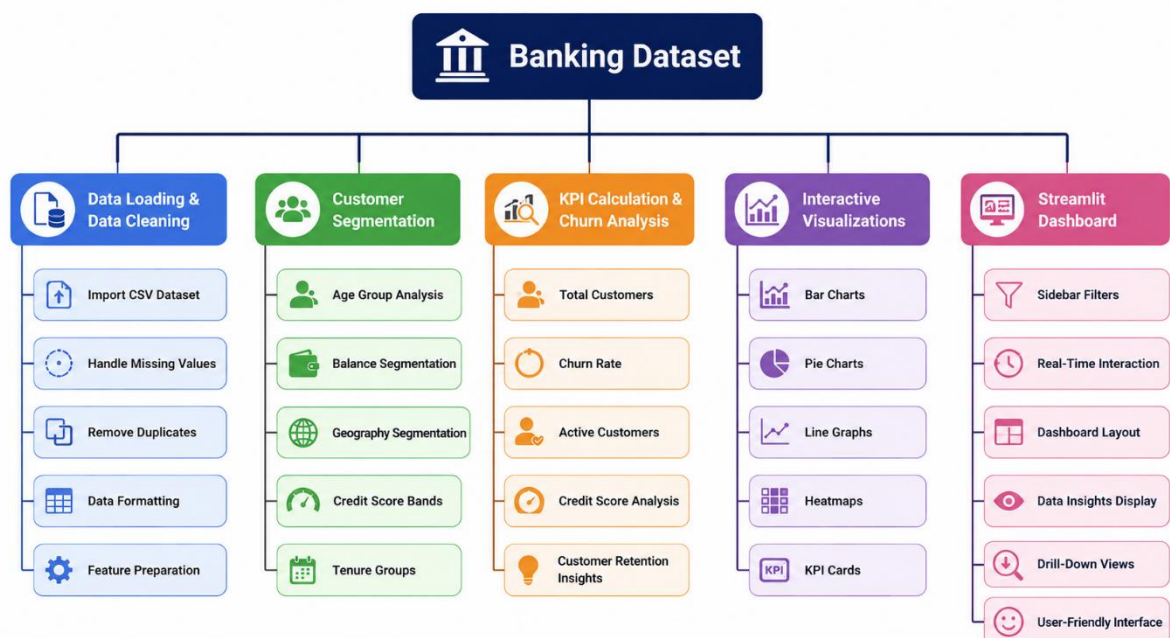
2.1 Flow of the project

This project follows a structured analytical workflow to transform raw banking data into meaning business insights through an interactive dashboard. The process begins with collecting and loading the cleaned customer dataset into the Python environment using the Pandas library. The dataset contains customer information such as geography, age, credit score, balance, tenure, salary, and activity status.

After loading the data, filtering and segmentation techniques are applied to categorize customers into different groups such as age groups, balance segments, credit score segments, and geography segments. The project then calculates important Key Performance Indicator (KPIs) including Overall churn rate, High-value customer churn ratio, Geographic risk index, and Engagement drop indicator.

Next, data visualization techniques are implemented using Plotly to generate interactive charts and graphs for churn analysis. Finally, the Streamlit framework is used to develop an interactive dashboard where users can apply filters, explore customer segments, analyze churn patterns, and perform drill-down analysis for better decision-making.

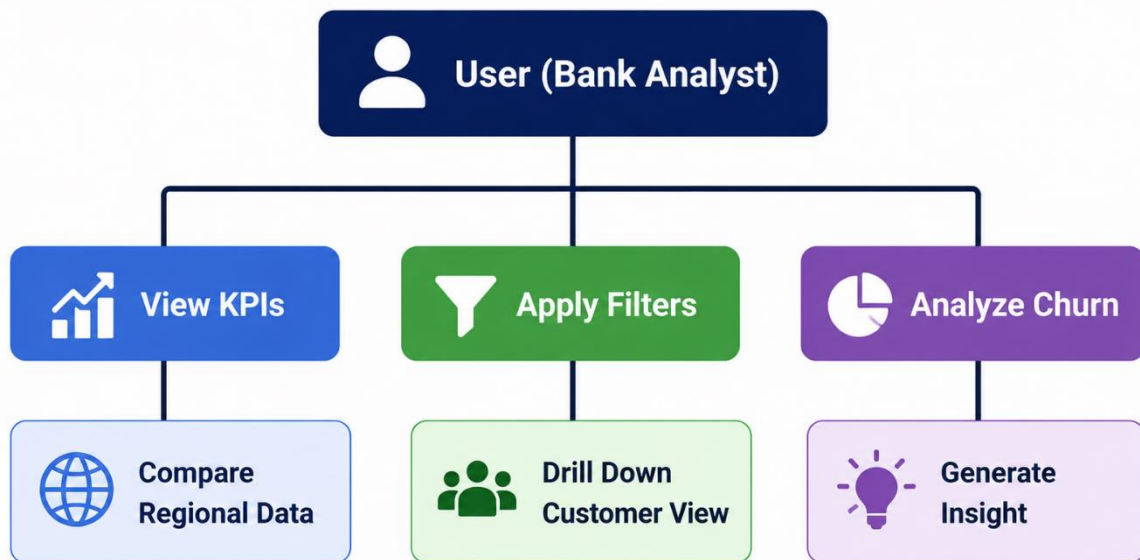
Flow of the project diagram



2.2 Use Case Diagram

The use case diagram represents how the user interacts with the churn analytics dashboard. The primary user of the system is the banking analyst or decision-maker who uses the dashboard to analyze customer churn patterns and customer segmentation.

Use case diagram are as follows:



2.3 Language and Platform Used

The project is developed using the **Python Programming language**, which is widely used in data analytics, machine learning, and dashboard development due to its simplicity and powerful libraries. Python libraries such as **Pandas** and **NumPy** are used for data processing and analysis, while **Plotly** and **Matplotlib** is used for creating interactive visualizations and charts.

```

import streamlit as st
import pandas as pd
import numpy as np
import plotly.express as px
import plotly.graph_objects as go
  
```

The project uses **the Streamlit platform** to build the interactive web dashboard. Streamlit allows quick development of professional dashboards with user-friendly interfaces and dynamic filtering options. The dashboard provides real-time interaction with customer data and helps users explore churn patterns efficiently.

```
import streamlit as st  
  
pip install streamlit  
  
streamlit run dashboard.py
```

Additionally, the project can be executed using development platforms such as **Visual Studio Code (VC Code), Jupyter Notebook, or PyCharm**, making it flexible and easy to manage for data analytics applications.

IMPLEMENTATION

3.1 Gathering Requirements and Defining Problem Statement

This is the first step wherein the requirements are collected from the clients to understand the deliverables and goals to be achieved after which a problem statement is defined which has to be adhered to while development of the project.

Problem Statement Explanation

Identifying high-risk customer segments

Banks often struggle to determine which groups of customers are most likely to leave their services. Without proper segmentation and analysis, it becomes difficult to create targeted retention strategies for high-risk customers.

Understanding churn differences by geography and demographics

Customer churn behavior varies across regions, age groups, and other demographics categories, making analysis complex for banking institutions. Banks need detailed insights to understand how geographic and demographic factors influence customer attrition.

Quantifying the financial profile of churned customers

Banks face challenges in evaluating the financial impact caused by customers who leave the organization. Identifying whether high-balance or high-value customers are churning is important for reducing revenue loss and improving profitability.

3.2 Data Collection and Importing

Data collection is a systematic approach for gathering and measuring information from a variety of sources in order to obtain a complete and accurate picture of an interest area. It helps an individual or organization to address specific questions, determine outcomes and forecast future probabilities and patterns.

This project is based on a structured banking customer dataset containing demographic, financial, and behavioral information of customers from European banks. The dataset was collected in CSV (Comma-Separated Values) format and includes important customer attributes such as age, geography, credit score, account balance, estimated salary, tenure, activity status, and customer churn information. These attributes are essential for analyzing customer behavior, identifying churn patterns, and performing customer segmentation.

Before implementing the dashboard, the dataset was cleaned and pre-processed to ensure data consistency and accuracy. Additional segmented columns such as **AgeGroup**, **CreditScoreSegment**, **TenureGroup**, **BalanceSegment**, and **GeographySegment** were created to support better analytical visualization and filtering within the dashboard. The cleaned

dataset was stored as “**European_churn_cleaned.csv**” and used as the primary data source for the project.

The importing process was implemented using the **Pandas** library in Python, which provides efficient functions for handling structured datasets. In the `dashboard.py` file, the dataset is imported using the `pd.read_csv()` function within a cached data-loading function. Streamlit’s `@st.cache` data decorator is used to improve dashboard performance by reducing repeated data loading operations during user interaction.

```
#-----LOAD DATA-----  
@st.cache_data  
def load_data():  
    df = pd.read_csv(r"European_churn_cleaned.csv")  
    return df  
df = load_data()
```

After importing the dataset, the data is stored in a `DataFrame` structure, enabling easy filtering, grouping, KPI calculation, and visualization generation. The imported dataset acts as the foundation for all dashboard functionalities, including customer segmentation, churn analysis, KPI generation, and interactive visual exploration.

Overall, the data collection and importing stage plays a crucial role in ensuring accurate analysis and smooth dashboard performance, making it the backbone of the entire customer churn analytics system.

3.3 Designing Databases

The database design for this project was structured to efficiently manage and analyze customer-related banking data. The project uses a cleaned CSV dataset as the primary data source, which acts as a relational data structure for performing customer segmentation, churn analysis, KPI calculations, and dashboard visualization.

The dataset was designed to include important customer attributes such as Customer ID, Geography, Age, Credit Score, Balance, Estimated Salary, Tenure, Activity Status, and Churn Status (Exited). To improve analytical performance and dashboard usability, additional segmented fields such as **AgeGroup**, **CreditScoreSegment**, **TenureGroup**, **BalanceSegment**, and **GeographySegment** were created during the preprocessing stage. These fields help organize customer information into meaningful categories for filtering and visualization.

The database structure supports easy grouping, filtering, and aggregation operations using the `Pandas DataFrame` framework in Python. The segmented database design allows the dashboard to dynamically calculate KPIs such as Overall Churn Rate, Geographic Risk Index, and High-Value Customer Churn Ratio.

The database was designed with a focus on simplicity, consistency, and analytical efficiency to ensure smooth integration with the Streamlit dashboard. This structured approach enables faster data processing, interactive customer analysis, and effective visualization generation for business intelligence and decision-making purposes.

3.4 Data Cleaning

Data cleaning is one of the most important stages in this project because the accuracy and reliability of the dashboard analysis depend on the quality of the dataset. Before performing customer segmentation, KPI generation, and churn analysis, the banking dataset was cleaned and preprocessed to remove inconsistencies, improve data quality, and prepare the data for visualization and analysis.

The raw banking dataset contained customer demographic, financial, and behavioral attributes such as geography, age, credit score, account balance, salary, tenure, activity status, and churn status. During the cleaning process, missing values, duplicate records, incorrect data formats, and inconsistent category labels were checked and corrected. This ensured that the dashboard calculations and visualizations produced accurate business insights.

One of the major tasks in the cleaning process was the creation of the segmented columns used in the dashboard filters and analysis. New categories such as **AgeGroup**, **CreditScoreSegment**, **TenureGroup**, **BalanceSegment**, and **GeographySegment** were generated to simplify customer segmentation and improve visualization clarity. These segmented columns allowed users to analyze churn patterns more effectively across different customer groups.

```
#CUSTOMER SEGMENTATION DESIGN
#geographic segment
df['GeographySegment'] = df.apply(lambda x: 'Germany'
    if x['Geography_Germany'] ==1
    else('Spain' if x['Geography_Spain']==1
    else 'France'), axis=1)

#age segmentation
df['AgeGroup'] = pd.cut(df['Age'],
    bins=[0, 30, 45, 60, 100],
    labels=['<30', '30-45', '45-60', '60+'])

#credit score segmentation
df['CreditScoreSegment'] = pd.cut(df['CreditScore'],
    bins=[300, 600, 700, 850],
    labels=['Low', 'Medium', 'High'])

#tenure segment
df['TenureGroup'] = pd.cut(df['Tenure'],
    bins=[-1, 3, 7, 10],
    labels=['New Customers', 'Mid-Term Customers', 'Long-Term Customers'])

#balance segment
df['BalanceSegment'] = pd.cut(df['Balance'],
    bins=[-1, 0, 100000, 300000],
    labels=['Zero-Balance', 'Low-Balance', 'High-Balance'])
```

Several important Python and Pandas functions were used during the data cleaning and preprocessing stage. The `pd.read_csv()` function was used to import the dataset into a Pandas DataFrame for structured analysis. Functions such as `df.isnull().sum()` and `df.drop()` helped identify and handle missing values and remove columns in the dataset. Duplicate records were checked using `df.duplicated()` and removed using `df.drop_duplicates()` to maintain data consistency.

```
d=pd.read_csv(r"C:\Users\HP\Downloads\European_Bank.csv")
```

```
#checking no. of nulls
print(df.isnull().sum())
```

```
# DATA CLEANING & PREPARATION
#removing column surname and year
df.drop(['Surname'], axis=1, inplace=True)
df.drop(['Year'], axis=1, inplace=True)
```

The `df.groupby()` function played an important role in organizing customer data into different segments for KPI calculations and churn analysis. Functions like `isin()` were used to apply dynamic filters within the Streamlit dashboard. Additionally, conditionally filtering techniques were used to identify premium customers, inactive customers, churned customers, and retained customers.

```
# 4] comparision of churned vs retained profiles
df.groupby('Exited').mean(numeric_only= True)
```

For segmentation purposes, techniques such as binning and categorization were applied to numerical columns like age, credit score, tenure, and balance. These transformations improved readability and supported better analytical visualization in the dashboard.

The cleaned dataset was finally stored as “**European_churn_cleaned.csv**”, which became the primary data source for the Streamlit dashboard. Overall, the data cleaning process improved the reliability, efficiency, and analytical quality of the project, enabling accurate customer churn analysis and interactive business intelligence reporting.

```
#df.to_csv
df.to_csv("European_churn_cleaned.csv", index = False)
```

3.5 Data Filtering

Data filtering is the method of choosing a smaller portion of the dataset and using that subset to view, analyze and evaluate data. Generally, filtering is temporary – the entire dataset is retained, but only part of it is used for calculation. It is also called subsetting or drill down data wherein data is extracted with respect to certain defined logical conditions.

Filtering is used for the following tasks:

- Analyzing results for a particular period of time
- Calculating results for particular groups of interest
- Exclude erroneous or “bad” observations from an analysis
- Train and validate statistical models

This project was designed to provide dynamic and user-focused analysis through an interactive dashboard environment. Using Python and Streamlit, multiple filtering options were implemented to allow users to explore customer behavior patterns efficiently across different banking segments. The filtering process enabled analysts to narrow down data based on important attributes such as geography, age groups, tenure, credit score bands, account balance ranges, product usage, active membership status, and churn category.

The dashboard applied real-time filtering techniques using Pandas and streamlit widgets, ensuring that all KPIs, charts, and visualizations updated instantly according to user selections. This improved the accuracy of churn analysis by allowing focused examination of specific customer groups and regional trends. Filters were also integrated with customer segmentation logic, making it easier to compare high-risk and low-risk customer profiles.

Additional filtering functionalities helped identify patterns related to customer retention, inactive users, and high-value customers likely to churn. The implementation enhanced data exploration, improved decision-making efficiency, and provided banking analysts with a flexible analytical environment for understanding customer behavior and predicting churn trends more effectively.

```
#apply filters
filtered_df = df[
    (df['GeographySegment'].isin(geo_filter)) &
    (df['AgeGroup'].isin(age_filter)) &
    (df['CreditScoreSegment'].isin(cr_score_filter)) &
    (df['TenureGroup'].isin(tenure_filter)) &
    (df['BalanceSegment'].isin(balance_filter))
]
```


3.6 Prototyping – Streamlit Dashboard

A prototype is an early version, model, or release of a product that is constructed to test a design or process. It is generally used by system analysts and users to assess a new design to enhance precision. Prototyping serves to specify a real, working system rather than a theoretical one. Creation of a prototype in some design workflow models is the step between formalizing and testing an idea.

The prototyping phase of the project focused on designing an interactive and user-friendly and analytical dashboard using the Python-based Streamlit framework. The primary objective of the prototype was to create a centralized platform where banking analysts could monitor customer behavior, evaluate churn trends, and perform customer segmentation through real-time data exploration and visualization.

The Streamlit dashboard was structured using a wide-layout interface to improve readability and support multiple analytical components on a single page. During the prototyping process, careful attention was given to the arrangements of KPIs, filters, visualizations, and drill-down sections to ensure smooth navigation and efficient analytical workflows. The sidebar filtering system was designed as the core interactive component of the dashboard, allowing users to dynamically filter customer data based on Geography Segment, Age Group, Credit Score Segment, Tenure Group, and Balance Segment. These filters enabled banking analysts to explore churn behavior across different customer categories and compare segment-wise performance effectively.

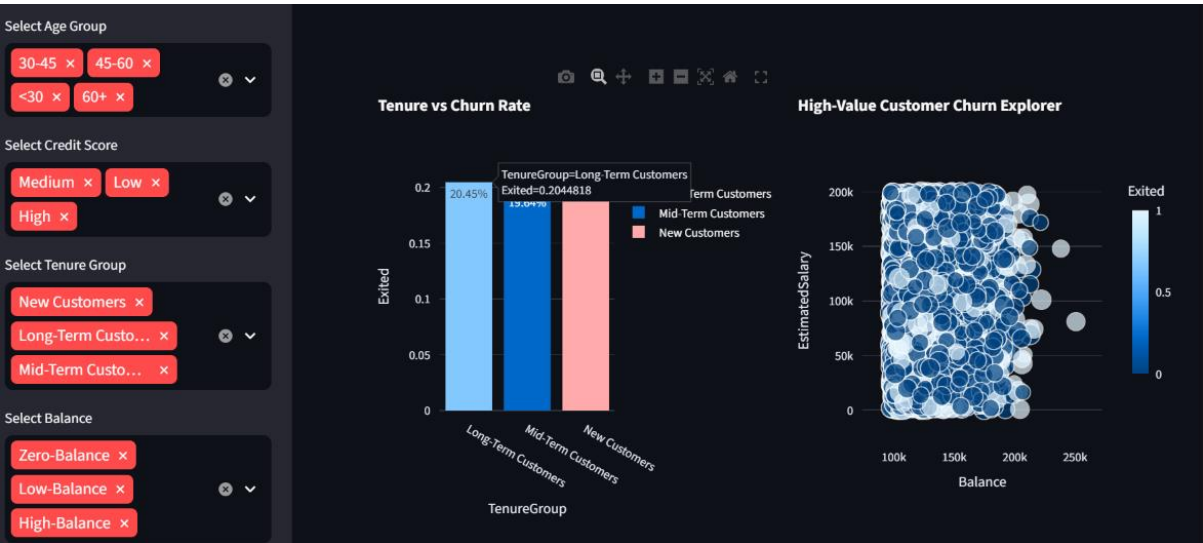
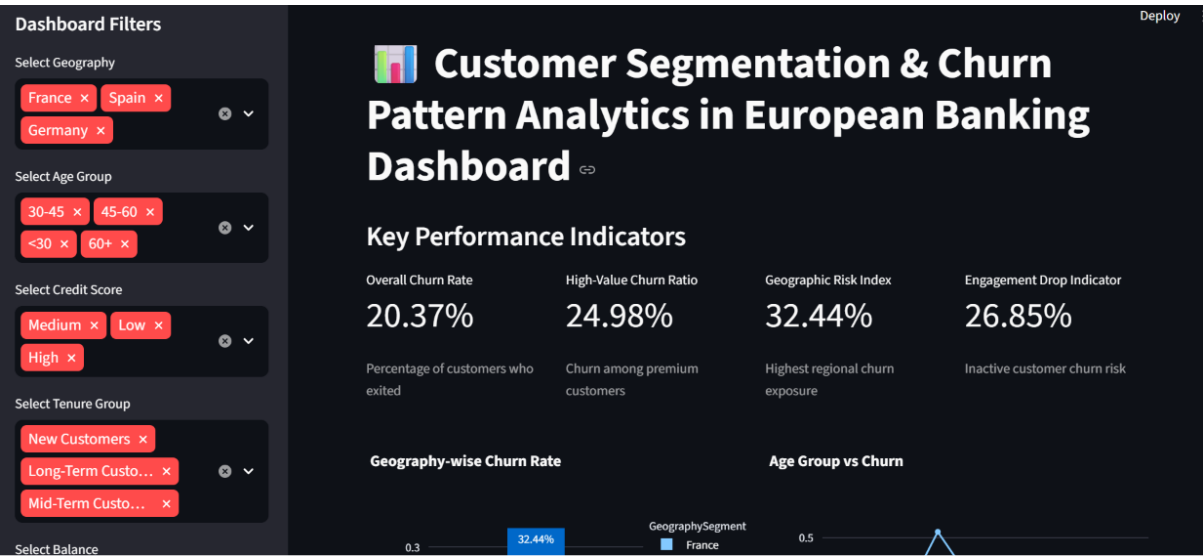
The dashboard prototype also incorporated several key performance indicators (KPIs) to provide quick business insights. Metrics such as Overall Churn Rate, High-Value Churn Ratio, Geography Risk Index, and Engagement Drop Indicator were developed to help identify customer retention risks and regional churn exposure. These KPIs were dynamically connected with the filtering system, ensuring that all metrics updated automatically according to user selections.

To improve visual analytics, multiple interactive charts were implemented using Plotly within the Streamlit environment. Geography-wise churn analysis was represented using bar charts to compare regional churn rates, while line charts were used to analyze churn trends across age groups. Additional visualizations such as tenure-based churn comparison and high-value customer churn exploration were included to support deeper customer behavior analysis. The scatter plot visualization further enhanced the prototype by enabling analysts to examine relationships between customer balance, salary, credit score, and churn status.

The prototype also included a segment churn analysis table and a drill-down customer view section for detailed data inspection. The drill-down functionality allowed users to filter churned, retained, or all customers, making the dashboard more interactive and business-oriented. Streamlit's real-time rendering capabilities significantly improved dashboard responsiveness and provided a seamless analytical experience.

The overall prototyping process helped establish the structural design, analytical flow, and user interaction model of the dashboard before final implementation. It ensured that the final

system was scalable, visually organized, and capable of supporting banking analysts in identifying churn patterns, customer risk segments, and retention opportunities efficiently.



Exploratory Data Analysis

4.1 Defining Visuals

The visual design section of the project focused on creating clear, interactive, and business-oriented data visualizations to support customer churn analysis and segmentation. The dashboard visuals were developed using Plotly within the Streamlit framework to provide dynamic and real-time analytical insights for banking analysts. The visualization layer was designed to simplify complex customer data and convert it into understandable graphical representations for faster decision-making.

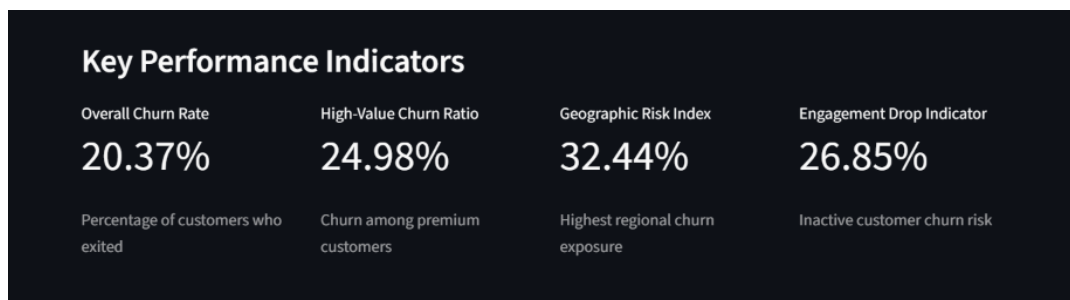
Several chart types were implemented to analyze customer churn patterns from multiple perspectives. Bar charts were used to compare geography-wise churn rates and tenure-based churn behavior, enabling users to identify high-risk customer regions and customer retention trends. A line chart was designed to visualize churn variation across different age groups, helping analysts understand age-related customer behavior patterns. Additionally, an interactive scatterplot was created to analyze high-value customers by comparing account balance, estimated salary, credit score, and churn status simultaneously.

The visuals were dynamically connected with dashboard filters, ensuring that all charts updated automatically according to user selections. This interactive visualization approach improved data exploration, enhanced analytical efficiency, and provided meaningful insights into customer retention risks and segmentation strategies within the European banking sector.

4.2 Visualizations

The visualization section of the project was developed to transform customer data into interactive and meaningful graphical insight for the banking analysts. Using Streamlit and Plotly, multiple visual components were integrated into dashboard to analyze customer segmentation, churn behavior, and regional performance effectively. The visualization design focused on improving business understanding through clear and dynamic data representation.

Key Performance Indicator (KPI)



1. Overall Churn Rate – 20.37%

- Approximately one out of every five customers has exited the bank, indicating a moderate customer retention challenge within the European banking sector.
- The churn percentage highlights the need for improved customer engagement and retention strategies.
- This metric serves as a primary indicator for monitoring customer loyalty and overall business stability.

2. High-Value Churn Ratio – 24.98%

- Premium customers with higher account balances show a significantly higher churn ratio compared to the average customer base.
- The finding suggests that high-value customers may be dissatisfied with service quality, engagement programs, or competitive banking offers.
- Retaining these customers is critical because they contribute heavily to revenue generation and long-term profitability.

3. Geographic Risk Index – 32.44%

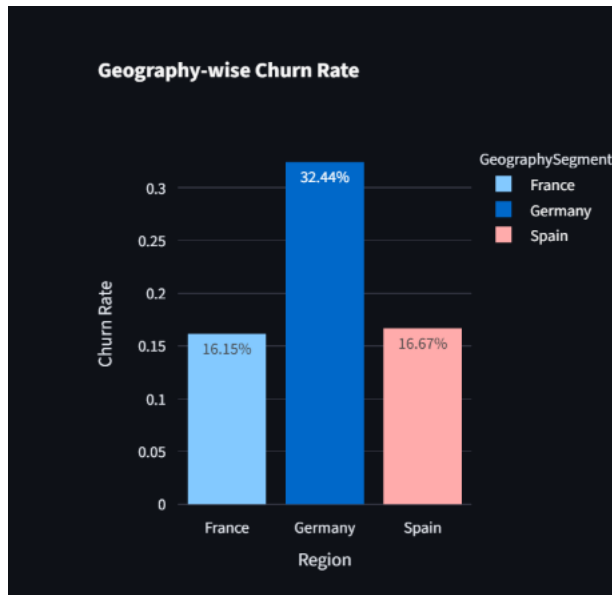
- Certain geographic regions demonstrate substantially higher churn exposure than others, indicating uneven customer satisfaction across locations.
- Regional churn variation may be influenced by local competition, service accessibility, economic conditions, or demographic behavior.
- This insight can help banking management prioritize region-specific customer retention campaigns.

4. Engagement Drop Indicator – 26.85%

- Inactive customers are more likely to churn compared to actively engaged customers.
- Reduced customer activity appears to be a strong indicator of future account closure or migration to competitors.
- The bank can reduce churn risk by introducing personalized engagement programs, loyalty offers, and digital banking incentives.

The KPI analysis revealed critical customer retention risks associated with inactive users, high-value customers, and region-specific churn exposure. These findings provide valuable business intelligence for improving customer engagement strategies, strengthening retention programs, and supporting data-driven decision-making within the European banking sector.

Geography -Wise Churn Rate Analysis



The above bar chart represents **the customer churn rate across different European regions**, specifically France, Germany, and Spain. The visualization compares the percentage of customers who exited the bank in each region, helping identify areas with higher customer retention risk.

Observations From The Chart

1. Germany shows the highest churn rate – 32.44%

- Germany recorded the highest churn rate among all regions, with approximately **32.44%** of customers leaving the bank.
- This indicates that nearly one-third of the customers in Germany discontinued banking services, making it the highest-risk region for customer attrition.
- The significantly high churn rate may be influenced by factors such as lower customer satisfaction, increased market competition, poor engagement levels, or changing customer preferences.
- Germany should be considered a priority region for customer retention programs and strategic business improvements.

2. France has the lowest churn rate – 16.15%

- France displayed the lowest churn rate at **16.15%**, including comparatively stronger customer retention and satisfaction levels.
- Customers in this region appear more stable and loyal to banking services.

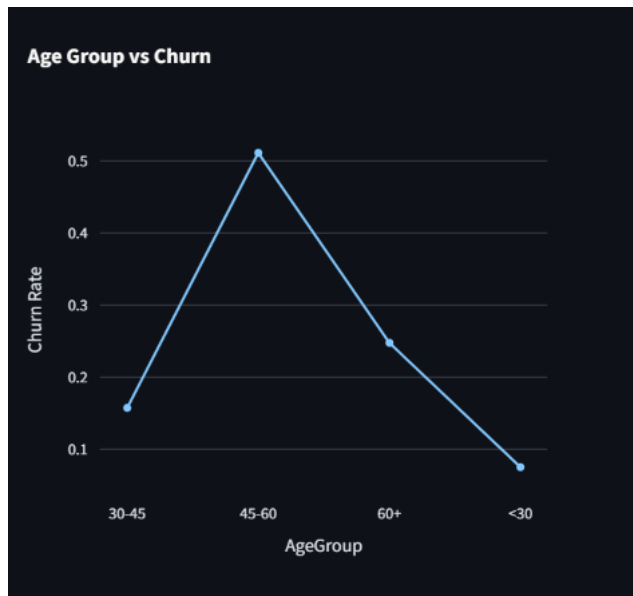
- The lower churn percentage suggests that existing banking strategies, customer engagement programs, or service quality in France may be more effective than in other regions.

3. Spain demonstrates moderate churn behavior – 16.67%

- Spain recorded a churn rate of **16.67%**, slightly higher than France but significantly lower than Germany.
- This indicates relatively balanced customer retention performance.
- While the churn level is not critically high, monitoring customer engagement and service satisfaction remains important to prevent future increases in churn.

The chart clearly highlights that customer churn behavior varies significantly across geographic regions. Germany represents the most vulnerable market segment, whereas France and Spain show comparatively stable customer retention patterns. These findings can help banking analysts and management teams develop region-specific retention strategies, improve customer engagement, and allocate business resources more effectively to reduce churn risk and enhance long-term customer loyalty.

Age Group Vs Churn



The above line chart represents the relationship between different customer age groups and their corresponding churn rates in the European banking sector. The visualization helps identify which age categories are more likely to discontinue banking services and highlights age-based customer retention patterns.

Observations From The Chart

1. Customers aged 45-60 show the highest churn rate

- The **45-60 age group** recorded the highest churn rate, reaching approximately **51%**.
- This indicates that more than half of the customers within this category are likely to leave the bank.
- The high churn level suggests that middle-aged customers may have higher financial expectations, stronger competition awareness, or increased sensitivity toward banking services and benefits.
- This age segment represents a critical risk group and requires targeted retention strategies, personalized services, and loyalty programs.

2. Customers above 60 years show moderate churn behavior

- The **60+ age group** displayed a churn rate of around **25%**.
- Although lower than the 45-60 category, the churn percentage still indicates a moderate retention challenge.
- Senior customers may face issues related to digital banking adaption, customer support expectations, or changing financial requirements.

- Improving customer assistance and simplified banking services could help reduce churn within this segment.

3. Customers aged 30-45 demonstrate stable retention

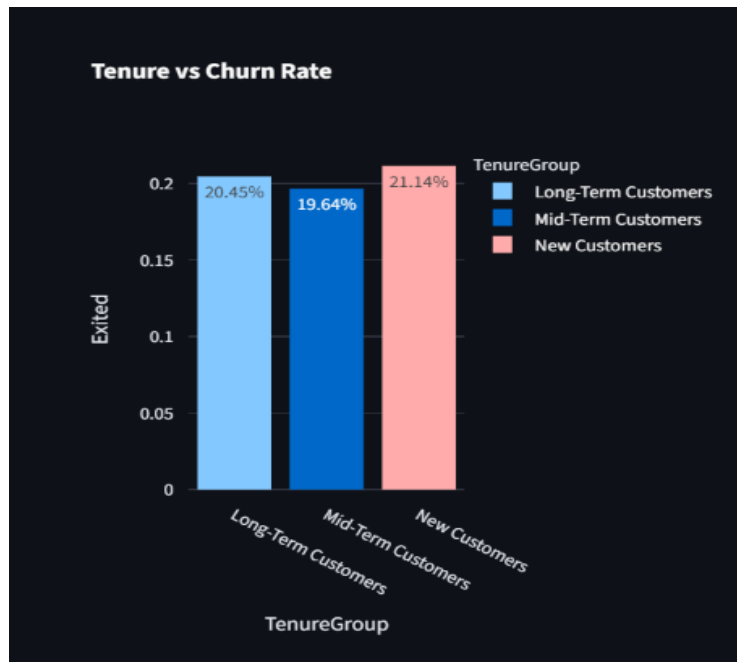
- The **30-45 age group** showed a churn rate of approximately **16%**, indicating relatively stable customer retention.
- Customers in this category appear more engaged and consistent in maintaining banking relationships.
- This segment may benefit from balanced financial products, active account usage, and long-term banking needs.

4. Customers below 30 have the lowest churn rate

- The **below 30 age group** recorded the lowest churn rate at nearly **8%**.
- Younger customers appear more likely to remain with the bank, possibly due to digital banking adoption, lower financial complexity, or satisfaction with online banking services.
- This group represents a strong opportunity for long-term customer relationship development.

The chart clearly demonstrates that customer churn behavior varies significantly across age groups. Middle-aged customers between 45-60 years represents the highest-risk segment, while younger customers shows stronger retention patterns. These findings can help banking organizations design age-specific customer engagement strategies, improve personalized financial services, and strengthened long-term customer loyalty programs.

Tenure Vs Churn Rate



The above bar chart illustrates the relationship between **customer tenure groups** and their **corresponding churn rates** in the European banking sector. The visualization compares churn behavior among New Customers, Mid-Term Customers, and Long-Term Customers to understand how customer loyalty duration affects retention patterns.

Observations From the Chart

1. New customers show the highest churn rate – 21.14%

- The New Customers category recorded the highest churn rate at approximately **21.14%**.
- This indicates that newly acquired customers are more likely to leave the bank during the early stages of their banking relationship.
- The higher churn percentage may result from lower customer engagement, unmet service expectations, lack of loyalty programs, or strong competition from other banks.
- This finding highlights the importance of onboarding strategies, customer support, and early engagement initiatives to improve retention among newly joined customers.

2. Long-term customers demonstrate moderate churn – 20.45%

- Long-Term Customers showed a churn rate of around **20.45%**, slightly lower than new customers.

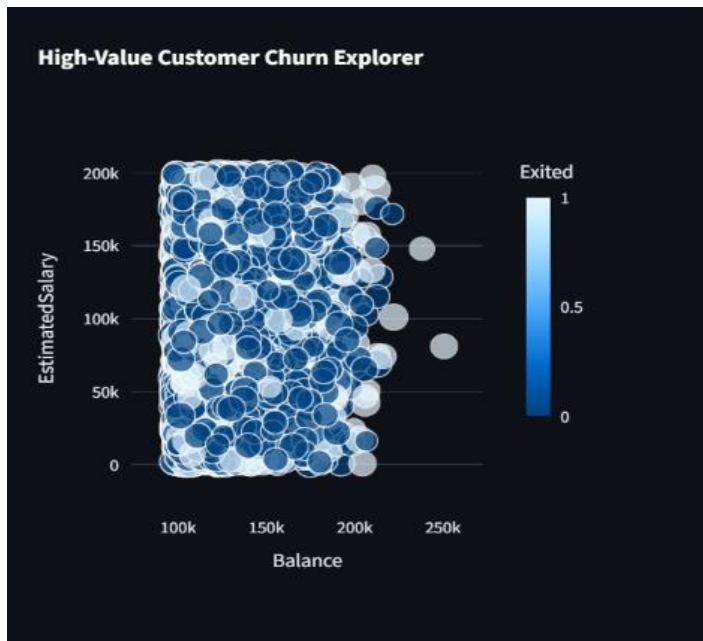
- Although these customers have maintained a longer relationship with the bank, the churn percentage suggests that customer loyalty alone does not completely eliminate churn risk.
- Long-Term customers may leave due to dissatisfaction with service quality, changing financial needs, or better competitive offers.
- Retention programs and personalized financial services remain important even for loyal customer segments.

3. Mid-term customers have the lowest churn rate – 19.64%

- The Mid-Term Customers group recorded the lowest churn rate at approximately **19.64%**.
- This suggests that customers in the middle stage of their banking relationship are relatively more stable and engaged.
- These customers may have developed trust in the bank while still actively using banking products and services.
- This segment appears to represent the most balanced retention group among all tenure categories.

The chart indicates that customer churn exists across all tenure groups, but new customers are slightly more vulnerable to leaving the bank. The relatively close churn percentages across the three categories suggest that churn is influenced not only by customer tenure but also by factors such as engagement level, service satisfaction, account activity, and financial product experience. These insights can help banking organizations design tenure-specific retention strategies, strengthen customer onboarding processes, improve long-term customer relationship management.

High-Value Customer Churn Explorer



The above scatter plot visualizes the relationship between **customer account balance**, **estimated salary**, **credit score size**, and **customer churn status** for high-value banking customers. This visualization was designed to explore churn behavior among premium customers and identify patterns associated with customer attrition in the European banking sector.

In the chart:

- The **X-axis** represents customer account balance.
- The **Y-axis** represents estimated salary.
- The **bubble size** indicates customer credit score levels.
- The **colour intensity** represents churn status, where darker and lighter shades indicate retained and churned customers respectively.

Observations From the Charts

1. High-value customers exist across all salary levels

- Customers with high account balances are distributed across a wide range of salary categories, from lower income levels to high-income groups.
- This indicates that customer balance is not solely dependent on salary and may also be influenced by investments, savings behavior, or long-term banking relationships.

2. Churn is present among premium customers

- The visualization clearly shows that both retained and churned customers are spread throughout the chart.
- Several high-balance customers have exited the bank despite maintaining strong financial profiles.
- This suggests that premium customers are also vulnerable to churn and cannot be retained through financial value alone.

3. Higher balance does not guarantee customer retention

- Customers with balances exceeding **200K** are still observed within churned categories.
- This finding indicates that high-value customers may leave due to service dissatisfaction, lack of personalized banking experience, competitive offers, or reduced engagement.

4. Credit score variation across churned customers

- Bubble sizes vary throughout the chart, showing that customers with both high and moderate credit scores are involved in churn behavior.
- This suggests that credit score alone is not a complete predictor of customer retention.

5. Dense customer distribution indicates strong data concentration

- Most customers are concentrated between balance ranges of approximately **80K-200K** and salary ranges of **0-200K**.
- This dense distribution represents the bank's primary high-value customer segment and highlights the importance of monitoring this group closely.

The scatter plot demonstrates that customer churn among high-value banking clients is influenced by multiple factors beyond salary and account balance. Even financially strong customers are at risk of leaving the bank, emphasizing the need for personalized engagement strategies, premium customer relationship management, and targeted retention programs. These insights can help banking organizations identify vulnerable high-value customers and improve long-term customer loyalty and satisfaction.

Segment Churn Rate Analysis

Segment Churn Rate Analysis			
	GeographySegment	Gender_Male	Churn Rate (%)
0	France	☐	20.34
1	France	☒	12.71
2	Germany	☐	37.55
3	Germany	☒	27.81
4	Spain	☐	21.21
5	Spain	☒	13.11

The above table presents a comparative analysis of customer churn rates based on **geographic region** and **gender segmentation** within the European banking sector. The analysis help identify how churn behavior differs across male and female customers in France, Germany, and Spain. This segmentation approach provides deeper business insights into customer retention patterns and demographic risk factors.

In the table:

- The **geography Segment** columns represent the customer's region.
- The **Gender_Male** column distinguishes male and female customer categories.
- The **Churn Rate (%)** column displays the percentage of customers who exited the bank within each segment.

Observations From the Chart:

1. Germany shows the highest overall churn across both genders

- Germany recorded the highest churn rates among all regions for both male and female customer groups.
- Female customers in Germany showed the highest churn rate overall at approximately **37.55%**.
- Male customers in Germany also demonstrated a high churn rate of around **27.81%**.
- These findings indicate that customer retention challenges are significantly stronger in Germany compared to France and Spain.

2. Female customers have higher churn rates than male customers

- Across all three regions, female customers consistently showed higher churn percentages than male customers.

- In France, female churn reached **20.34%**, while the male churn was approximately **12.71%**.
- In Spain, female churn was around **21.21%**, compared to **13.11%** for male customers.
- This trend suggests that female customers may be more sensitive to service quality, customer engagement, or competitive banking offers.

3. France and Spain show more stable customer retention

- France and Spain displayed relatively **lower churn rates** compared to Germany.
- Male customers in both countries demonstrated stronger retention behavior with **churn rates below 15%**.
- These regions appear to maintain comparatively stable customer satisfaction and engagement levels.

4. Gender-based retention patterns are clearly visible

- The segmentation analysis highlights that churn behavior is not only influenced by geography but also by demographic characteristics such as gender.
- The noticeable difference between male and female churn rates suggests that customer experience and retention strategies may affect customer groups differently.

The segment churn analysis reveals that Germany represents the highest -risk region for customer attrition, particularly among female customers. Additionally, female customers across all regions show higher churn tendencies compared to male customers. These findings emphasize the importance of implementing region-specific and demographic-focused retention strategies. Banking organizations can use these insights to design personalized customer engagement programs, improve service quality, and develop targeted loyalty initiatives to reduce churn and strengthen long-term customer relationships.

Drill-Down Customer View Analysis

Drill-Down Customer View

Select Customer Status

Churned

All

Churned

Retained

Drill-Down Customer View

Select Customer Status

All

	CreditScore	Age	Tenure	Balance	NumOfProducts	HasCrCard	IsActiveMember	EstimatedSalary	Exited	Ge
0	619	42	2	0	1	1	1	101348.88	1	
1	608	41	1	83807.86	1	0	1	112542.58	0	
2	502	42	8	159660.8	3	1	0	113931.57	1	
3	699	39	1	0	2	0	0	93826.63	0	
4	850	43	2	125510.82	1	1	1	79084.1	0	
5	645	44	8	113755.78	2	1	0	149756.71	1	
6	822	50	7	0	2	1	1	10062.8	0	
7	376	29	4	115046.74	4	1	0	119346.88	1	
8	501	44	4	142051.07	2	0	1	74940.5	0	

The above section represents **The Drill-Down Customer View** component of **The Customer Segmentation and Churn Pattern Analytics in European Banking Dashboard**. This feature was designed to provide detailed customer-level analysis by allowing users to dynamically filter and inspect customer records based on churn status. The Drill-Down functionality improves analytical depth by enabling banking analysts to move from high-level KPI insights to detailed customer information.

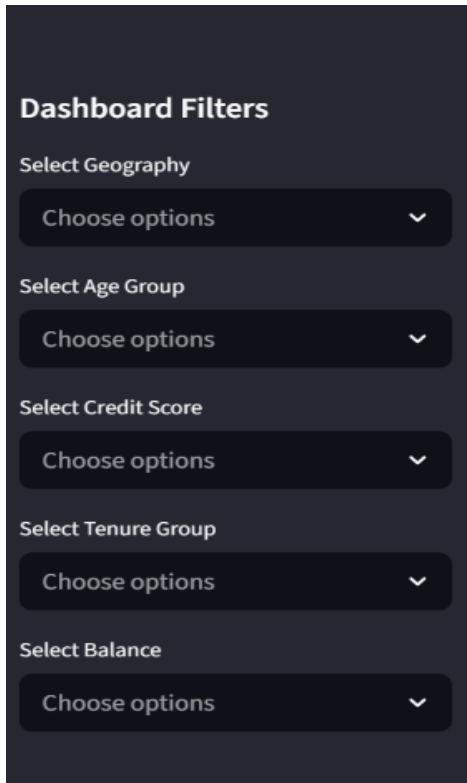
The dropdown filter contains three categories:

- **All Customers**
- **Churned Customers**
- **Retained Customers**

Based on the selected option, the dashboard dynamically updates the displayed customer records in real time.

The Drill-Down Customer View provides detailed operational insights into customer behavior, engagement, and churn patterns. The analysis reveals that churn is influenced by a combination of activity level, product usage, customer engagement, and services experience rather than a single financial factor. This functionality enhances business intelligence capabilities by supporting targeted customer retention strategies, personalized banking services, and data-driven decision-making within the European banking sector.

Dashboard Filters



The image shows a dark-themed sidebar titled "Dashboard Filters". It contains five sections, each with a label and a dropdown menu:

- Select Geography**: Choose options
- Select Age Group**: Choose options
- Select Credit Score**: Choose options
- Select Tenure Group**: Choose options
- Select Balance**: Choose options

The above interface represents the **Dashboard Filters** section of the **Customer Segmentation and Churn Pattern Analytics in European Banking Dashboard**. This component was developed using Streamlit sidebar widgets to provide interactive and dynamic data filtering capabilities for banking analysts. The filtering system serves as the primary control panel of the dashboard, allowing users to customize data exploration and perform targeted customer churn analysis.

The filter section includes five major customer segmentation categories:

- **Geography**
- **Age Group**
- **Credit Score**
- **Tenure Group**
- **Balance Segment**

Each dropdown filter enables users to specific customer categories according to analytical requirements. Once selections are applied all **KPIs, charts, tables** and **visualization** across the dashboard automatically update in real time.

The **Dashboard Filters** section significantly enhances the interactivity, flexibility, and analytical power of the banking dashboard. The filtering system allows users to perform

customized churn analysis across multiple customer dimensions, enabling deeper business insights and better decision-making. This feature supports efficient customer segmentation, targeted retention analysis, and dynamic exploration of customer behavior patterns within the European banking sector.

Conclusion And Future Scope

The customer segmentation and churn pattern analytics in European banking project successfully developed an interactive analytical dashboard to identify customer churn patterns, analyze customer segments, and support data-driven decision-making within the banking sector. By integrating Python, Streamlit, Pandas, and Plotly, the project transformed raw banking data into meaningful business insights through dynamic visualizations, KPI monitoring, and real-time filtering capabilities, the dashboard provided banking analysts with a centralized platform to monitor customer behavior and evaluate factors influencing customer retention and churn.

The project effectively implemented customer segmentation techniques using attributes such as geography, age group, credit score segment, tenure group, account balance, and activity status. These segmentation methods helped identify high-risk customer categories and allowed deeper analysis of churn behavior across different customer profiles. The filtering system improved analytical flexibility by enabling users to explore data interactively and customize insights according to specific business requirements.

Several KPIs, including Overall Churn Rate, High-Value Churn Ratio, Geographic Risk Index, and Engagement Drop Indicator, were developed to provide quick performance insights and highlights critical business risks. Interactive visualizations such as geography-wise churn comparison, age-based churn trends, tenure analysis, and high-value customer exploration enhanced the dashboard's ability to communicate complex banking data in a simplified and visually understandable manner.

The Streamlit-based dashboard also improved user interaction by supporting drill-down customer analysis and real-time updates. Analysts could examine churned and retained customer groups separately, making it easier to identify customer retention opportunities and understand behavioral patterns associated with churn. The use of Plotly visualizations further improved dashboard interactivity and analytical presentation quality.

Overall, the project demonstrated how data analytics and visualization tool can support strategic banking decision by identifying customer retention risks and improving segmentation analysis. The dashboard prototype provided a scalable and efficient analytical solution that can assist banking organizations in enhancing customer engagement, reducing churn rates, and optimizing retention strategies. The project also strengthened practical knowledge in data cleaning, segmentation, dashboard development, visualization design, and business analytics implementation using Python-based technologies.

REFERENCE

The following websites have been referred for Python and Streamlit tutorials:

- a. [Our Documentation | Python.org](#)
- b. [Streamlit documentation](#)
- c. [YouTube](#)
- d. [Wikipedia](#)
- e. [GitHub · Change is constant. GitHub keeps you ahead. · GitHub](#)